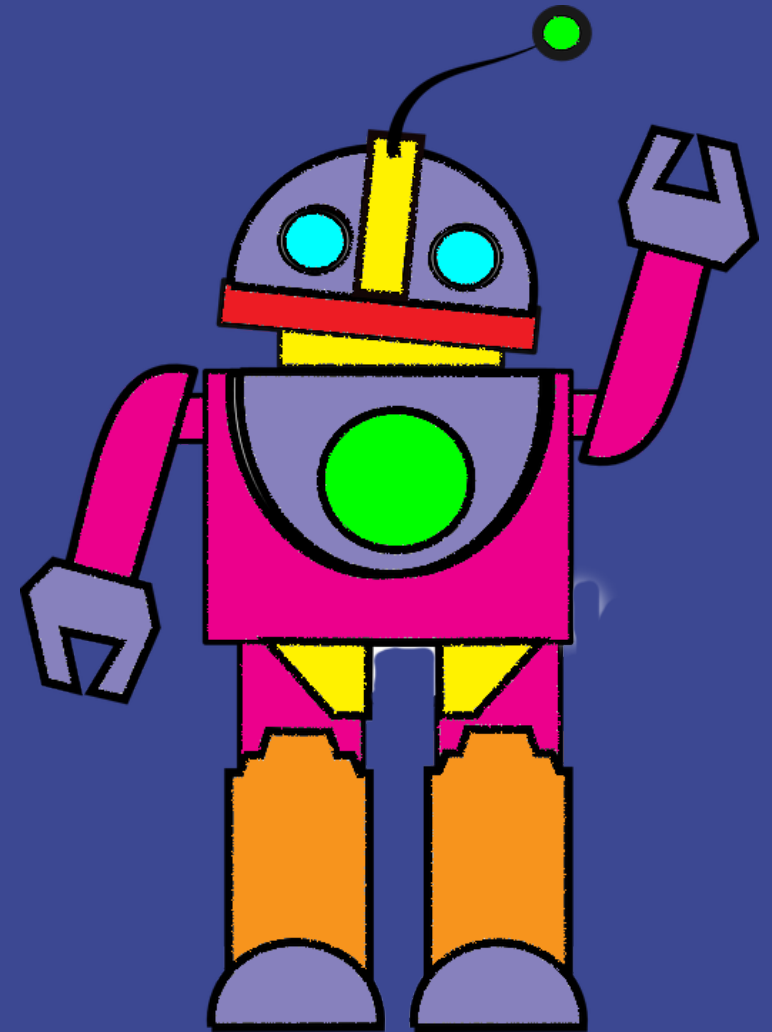


CS 156

Introduction to Artificial
Intelligence

Intelligent Agents



Today: Intelligent Agents

- ✓ Agents and environments
 - ▶ Rationality
 - ▶ Task environment: PEAS
 - ▶ Environment properties
 - ▶ Agent Types

Rational Agent

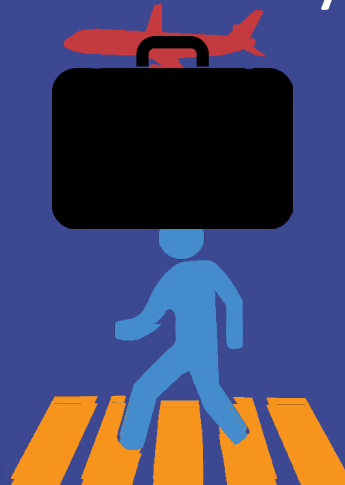
Rationality depends on:

- ▶ the performance measure
- ▶ the agent's knowledge of the environment
- ▶ the actions that the agent can perform
- ▶ the agent's percept sequence

A rational agent chooses the action that **maximizes the expected value** of the performance measure given the **percept sequence to date** and its knowledge of the environment.

Rationality - Limitations

- ▶ A rational agent is not omniscient: percepts may not supply all relevant information.
- ▶ A rational agent is not clairvoyant: percepts do not provide information about future events.
- ▶ A rational agent is not necessarily successful.



Rationality - Limitations

- ▶ A rational agent is not omniscient: percepts may not supply all relevant information.
- ▶ A rational agent is not clairvoyant: percepts do not provide information about future events.
- ▶ A rational agent is not necessarily successful.
- ▶ Difficulty: computational limitations may make perfect rationality unachievable.
- ▶ Design best program for given resources.

Rationality

Rationality involves:

- ▶ exploration
- ▶ learning
- ▶ autonomy

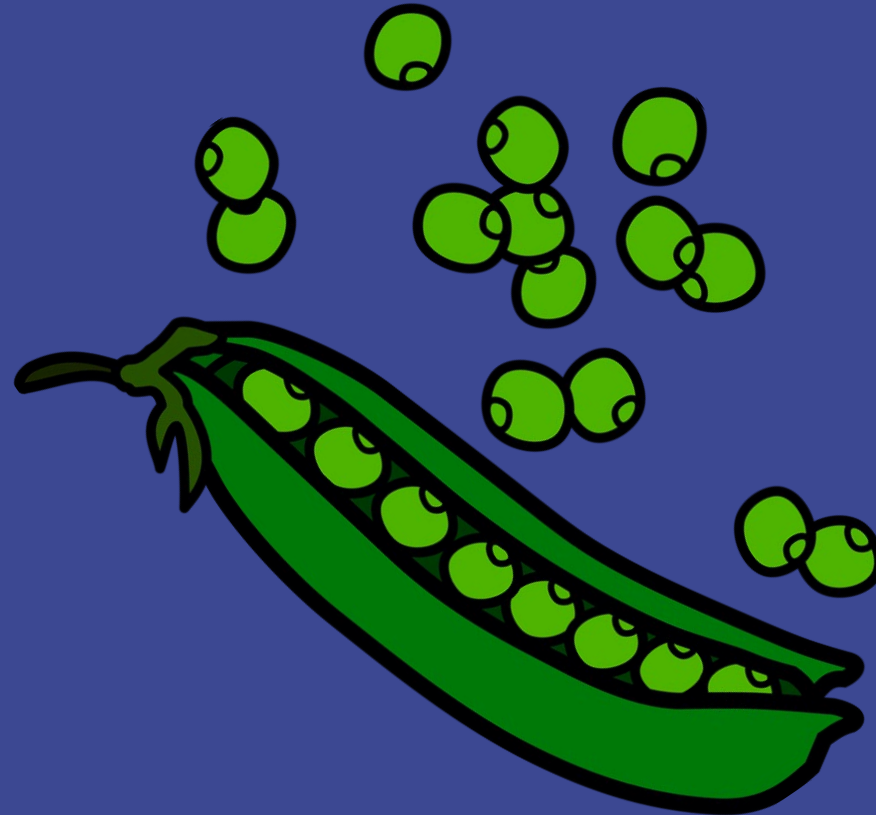
Intelligent Agents

- ✓ Agents and environments
- ✓ Rationality
- ▶ Task environment: PEAS
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Task Environment - PEAS

To design a rational agent, we must specify the task environment:

- ▶ Performance measure
- ▶ Environment
- ▶ Actuators
- ▶ Sensors



Task Environment - PEAS

Self-driving taxi

Performance measure: safety, destination, profits, legality, comfort

Environment: streets/freeways, traffic, pedestrians

Actuators: steering, accelerator, brake, horn, speaker/display

Sensors: camera, radar, speedometers, engine sensors, microphone, touchscreen, GPS



Task Environment - PEAS

Internet shopping agent



Performance measure: price, quality, appropriateness, efficiency

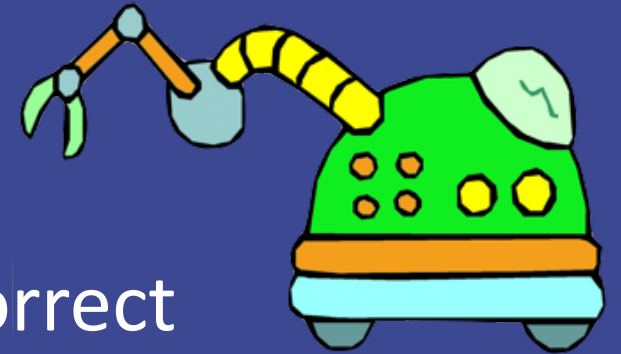
Environment: websites, vendors

Actuators: display to user, follow URL, fill in form

Sensors: keyboard, HTML pages (text, graphics, scripts)

Task Environment - PEAS

Part picking robot



Performance measure: percentage of parts in correct bins

Environment: conveyor belt with parts, bins

Actuators: jointed arm and hand

Sensors: camera

Intelligent Agents

- ✓ Agents and environments
- ✓ Rationality
- ✓ Task environment: PEAS
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Environment Properties



Environment Properties

Fully Observable vs Partially Observable

Fully observable: the agent's sensors give it access to the complete (relevant) state of the environment at each point in time.

- ▶ Chess?
- ▶ Automated taxi?
- ▶ The real world?

When the environment is partially observable, the agent needs to keep track of what it has seen of the environment so far.

Environment Properties

Deterministic vs Non-deterministic (Stochastic)

Deterministic: the next state of the environment is completely determined by the current state and the action executed by the agent.

- ▶ Chess?
- ▶ Automated taxi?
- ▶ The real world?

Environment Properties

Episodic vs Sequential

Episodic: The agent's experience is divided into **atomic episodes**. Each episode consists of the agent perceiving and then performing a single action. **The next episode does not depend on the actions taken in previous episodes.**

- ▶ Chess?
- ▶ Part picking robot?
- ▶ The real world?

Environment Properties

Static vs Dynamic

Static: the environment does not change **while an agent is deliberating**.

Semidynamic: the environment itself does not change with the passage of time but the agent's performance score does.

- ▶ Chess? (with without a clock?)
- ▶ Automated taxi?
- ▶ The real world?

Environment Properties

Discrete vs Continuous

Discrete: a **finite** number of distinct, clearly defined percepts and actions.

- ▶ Chess?
- ▶ Automated taxi?
- ▶ The real world?

Environment Properties

Single Agent vs Multiagent

Single agent: an agent operating by itself in an environment.

Multiagent environments may be **competitive** or **cooperative**.

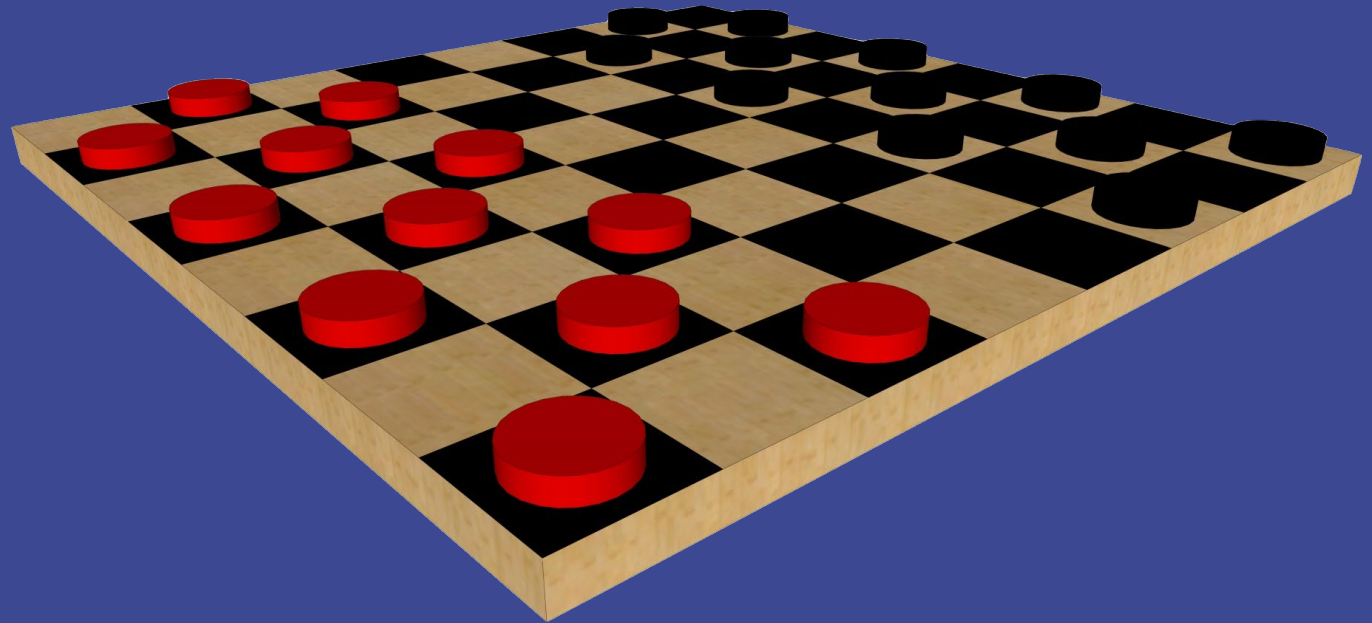
- ▶ Chess?
- ▶ Automated taxi?
- ▶ Part picking robot?
- ▶ The real world?

iClicker

Consider an agent playing checkers.

Which of the following is **NOT** a property of its environment?

- A. fully observable
- B. deterministic
- C. episodic
- D. static
- E. discrete



Backgammon?

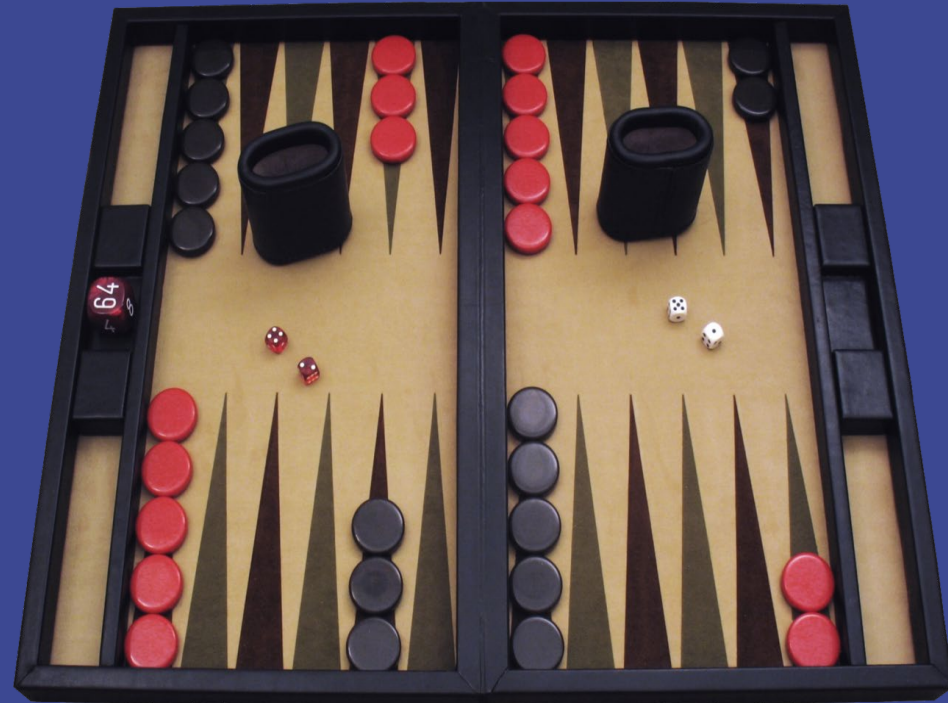


iClicker

Consider an agent playing backgammon.

Which of the following is a property of its environment?

- A. partially observable
- B. deterministic
- C. episodic
- D. discrete
- E. single-agent



iClicker

Consider an agent performing medical diagnosis.

Which of the following is a property of its environment?

- A. fully observable
- B. deterministic
- C. episodic
- D. dynamic
- E. discrete



Environment Properties

	Checkers	Backgammon	Medical Diagnosis
observable	Yes	Yes	No
deterministic	Yes	No	No
episodic	No	No	No
static	Yes	Yes	No
discrete	Yes	Yes	No
single agent	No	No	Yes

Intelligent Agents

- ✓ Agents and environments
- ✓ Rationality
- ✓ Task environment: PEAS
- ✓ Environment properties
- ▶ Agent Types

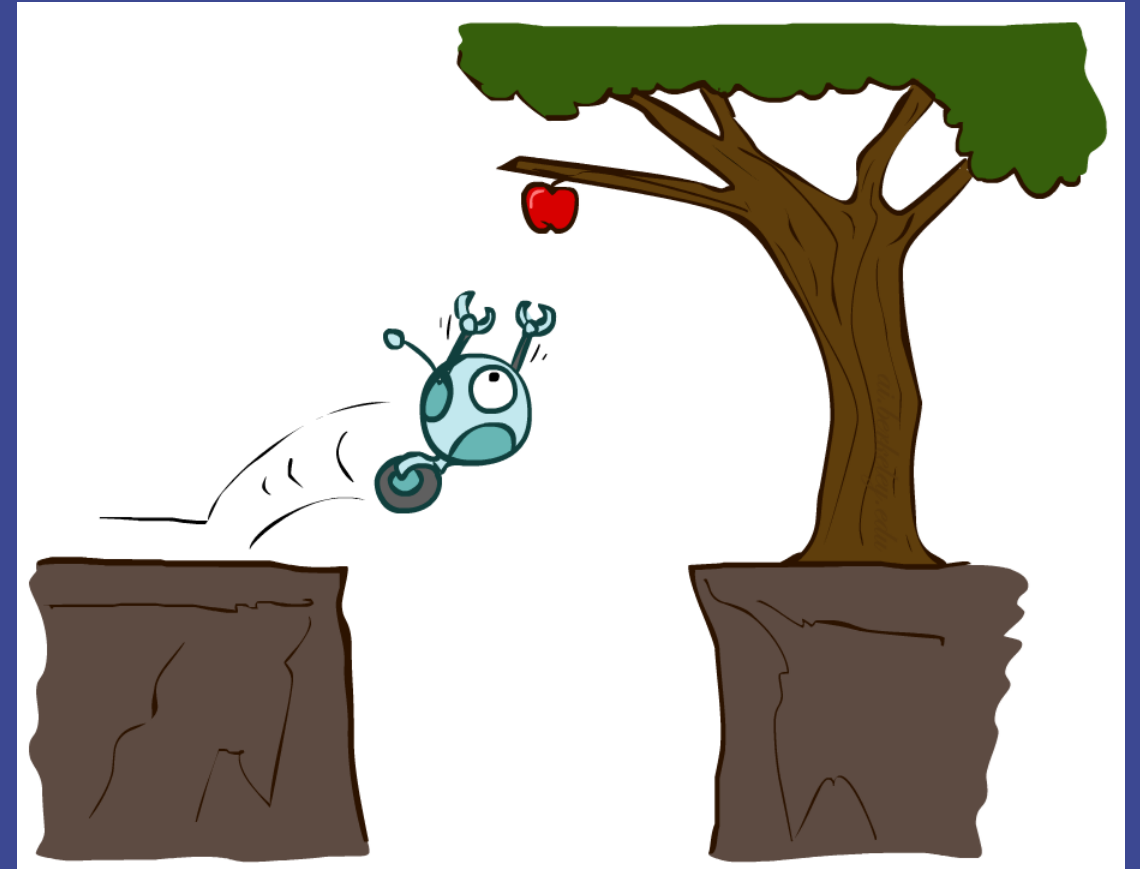
Agent Types

- ▶ Simple Reflex agents
- ▶ Model based agents:
 - reflex agents with state
 - goal-based agents
 - utility-based agents

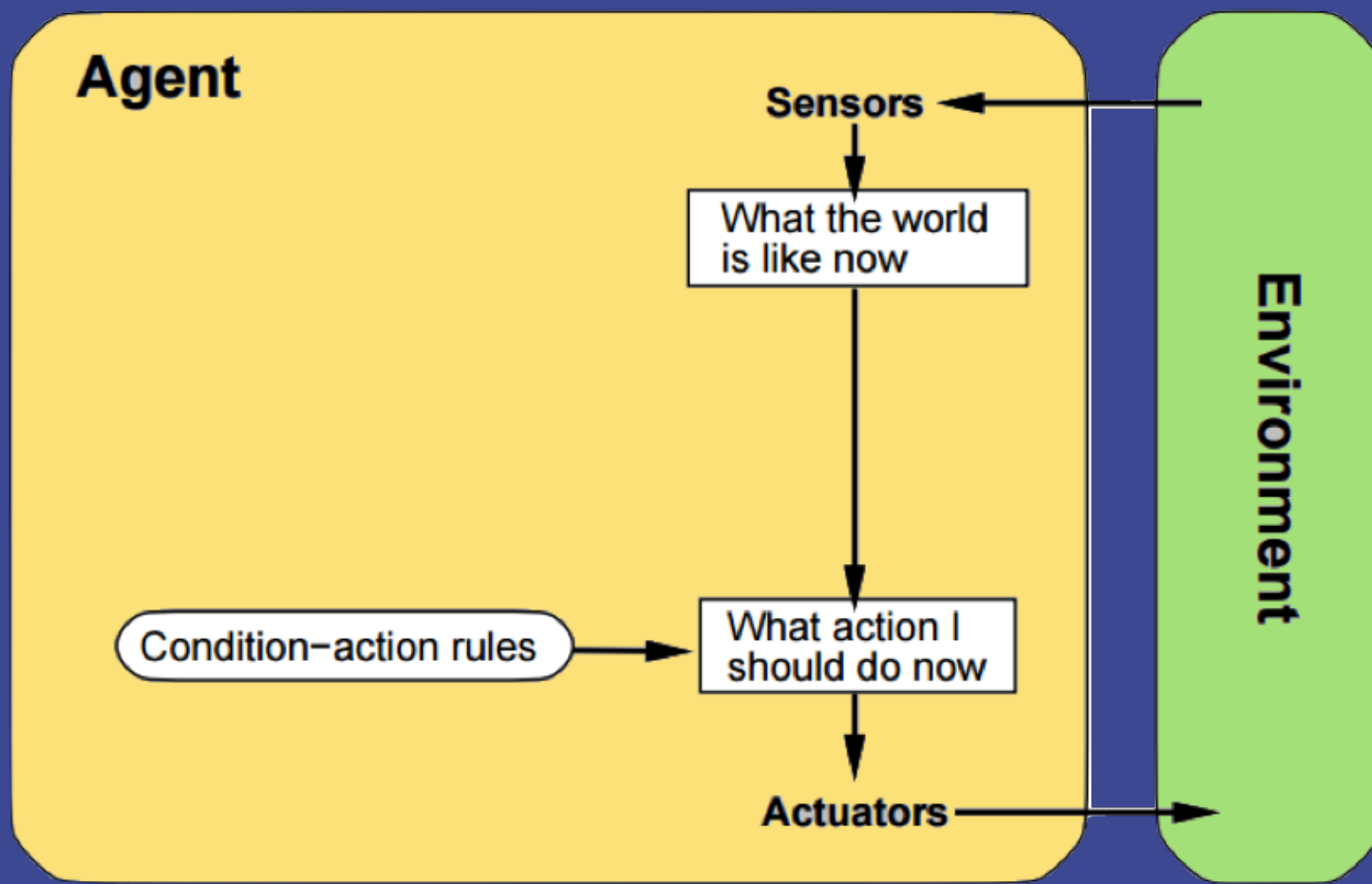
All these agents can be turned into **learning** agents

Simple Reflex Agents

- Choose action based on current percept
- Do not consider the future consequences of their actions
- Consider how the world IS

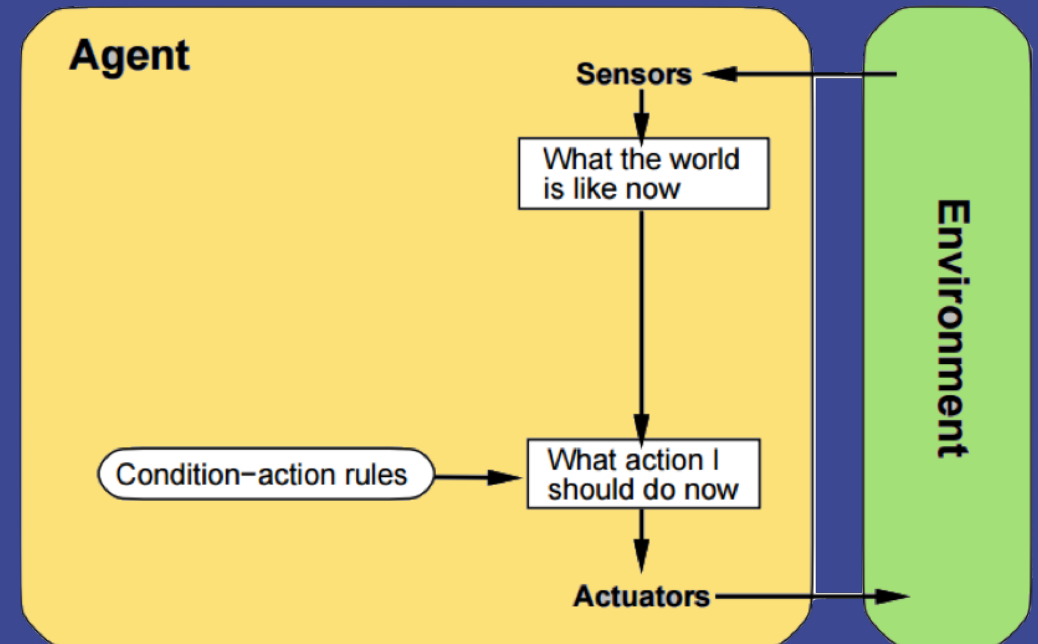
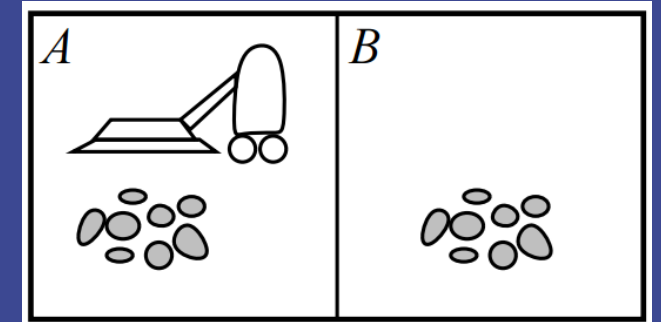


Simple Reflex Agent



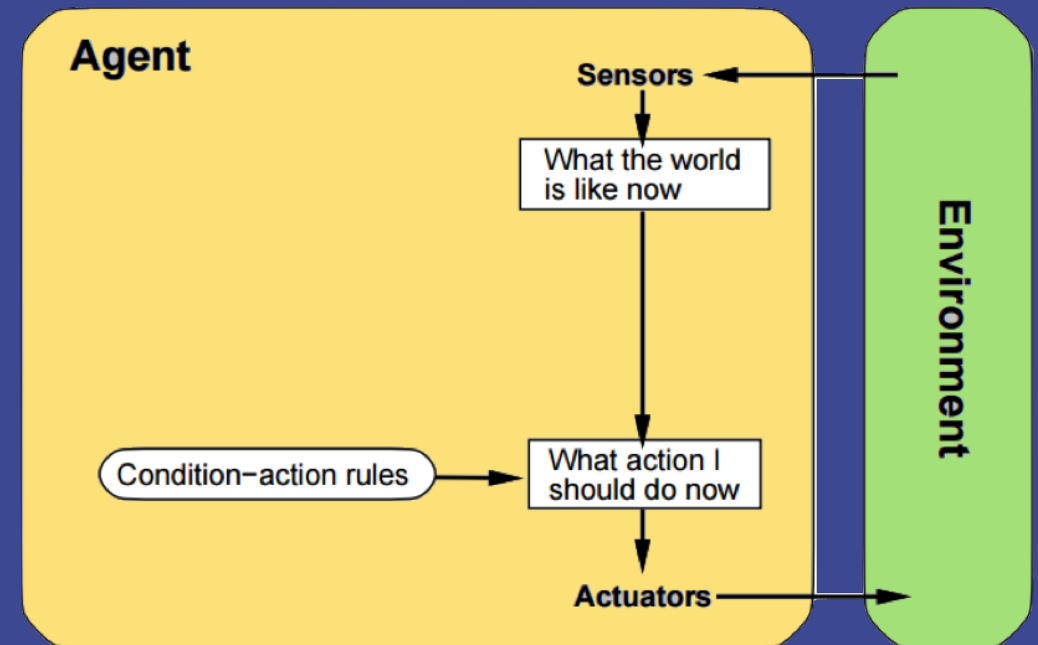
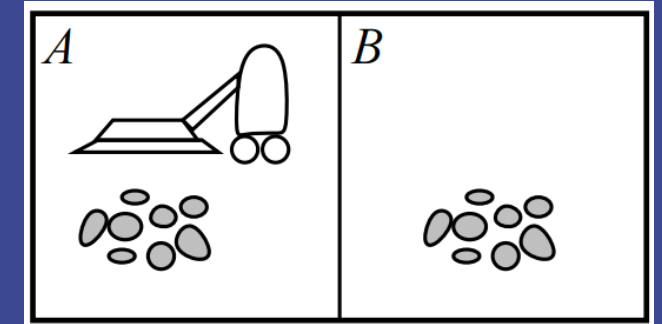
Simple Reflex Agent

- ▶ Choose action based on **current percept**
- ▶ Do not consider the future consequences of its actions
- ▶ Consider **how the world IS**



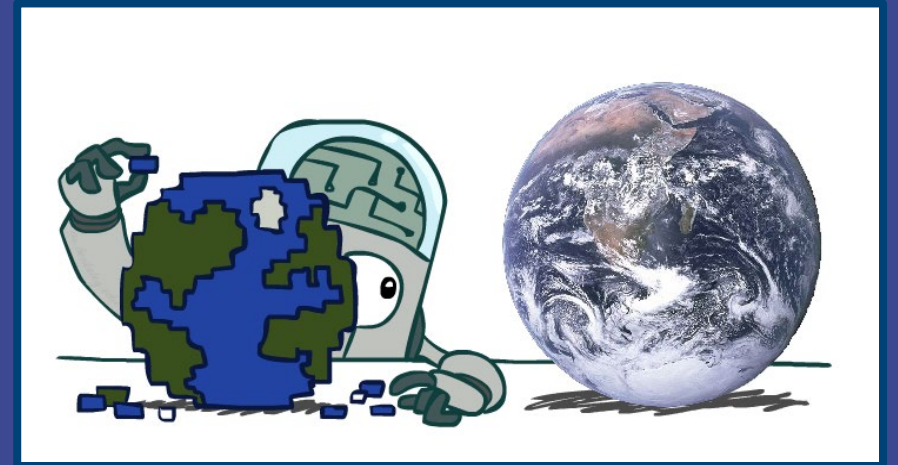
Simple Reflex Agent

```
def reflex_vacuum_agent(location, status):  
    if status == "Dirty":  
        return "Suck"  
    elif location == "A":  
        return "Right"  
    else: # location is B  
        return "Left"
```

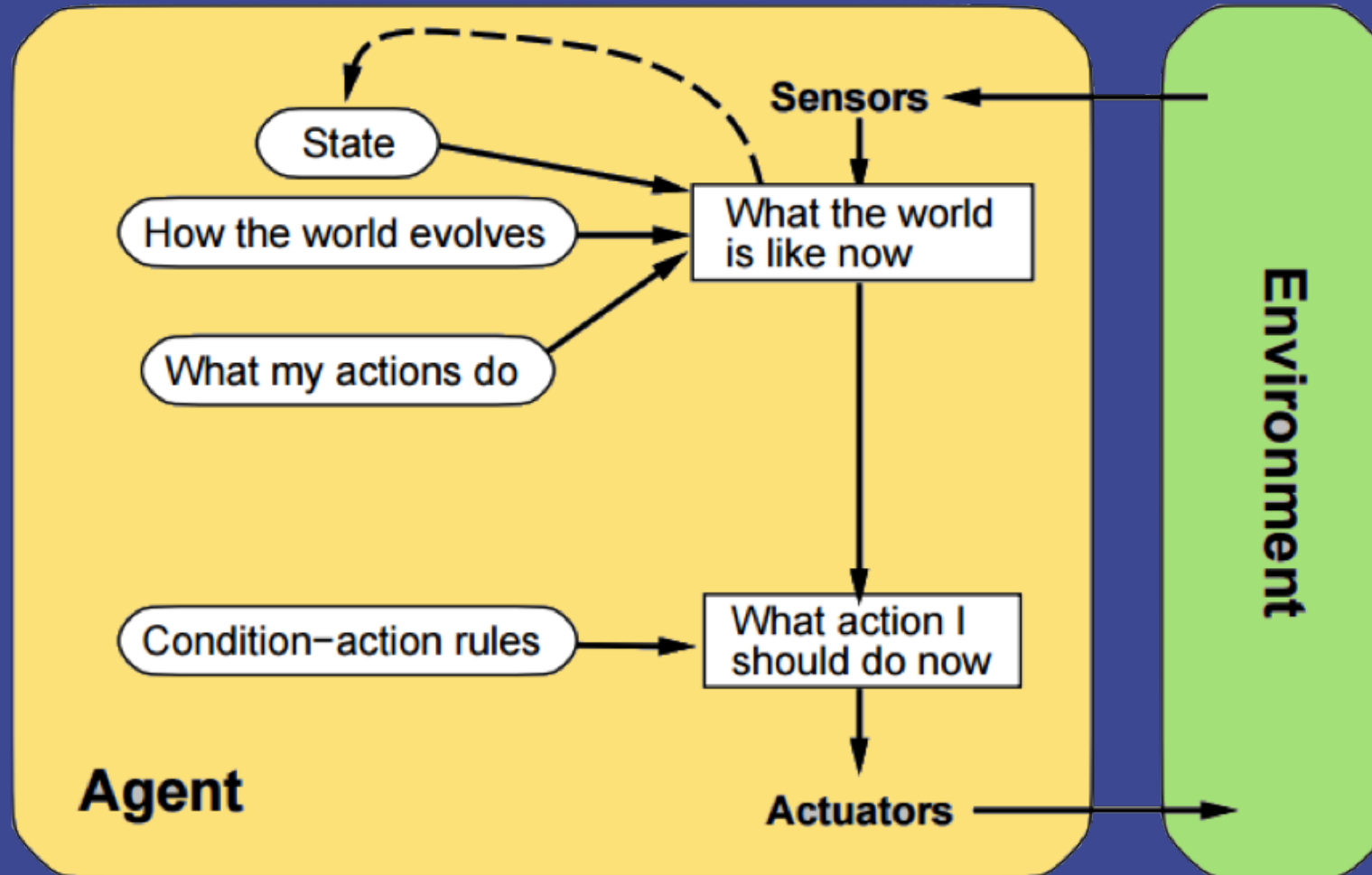


Model Based Agents

- ▶ Model based agents must have a model of how the world evolves (transition model):
 - reflex agents with state
 - goal-based agents
 - utility-based agents



Reflex Agent with State

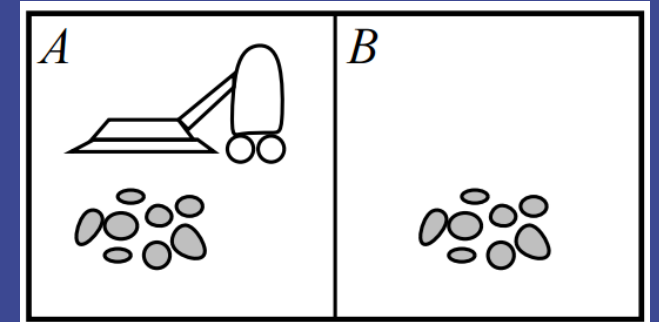


Vacuum Cleaner Agent

Is the environment

A. fully observable?

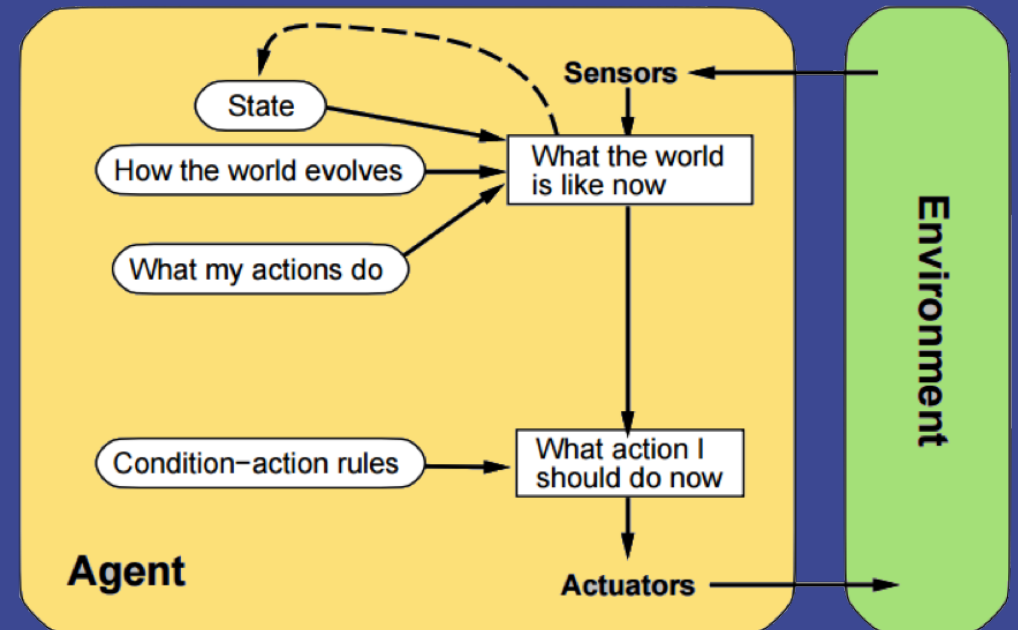
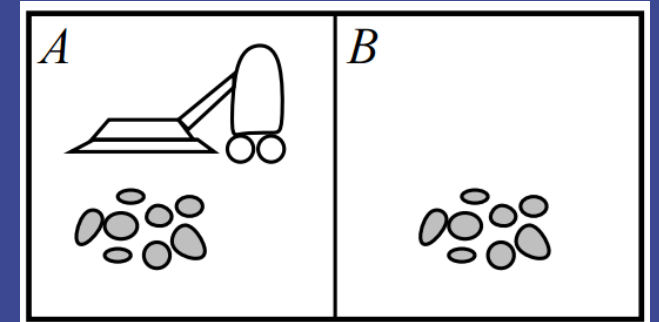
B. partially observable?



State/memory allows the agent to handle partially observable environments .

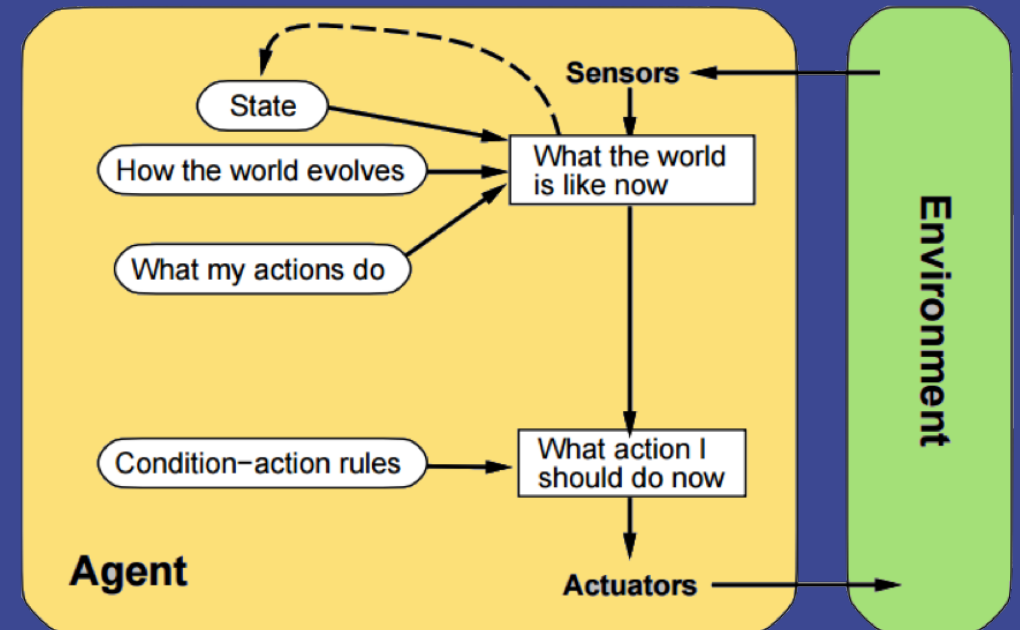
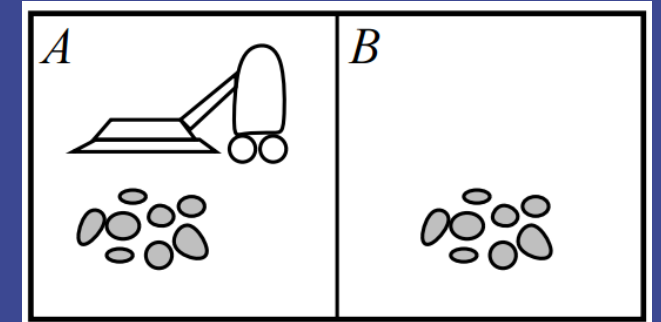
Reflex Agent with State

- ▶ Choose action based on current percept **and memory**
- ▶ May have memory and/or a model of the world's current state
- ▶ Do not consider the future consequences of their actions
- ▶ Consider how the world IS

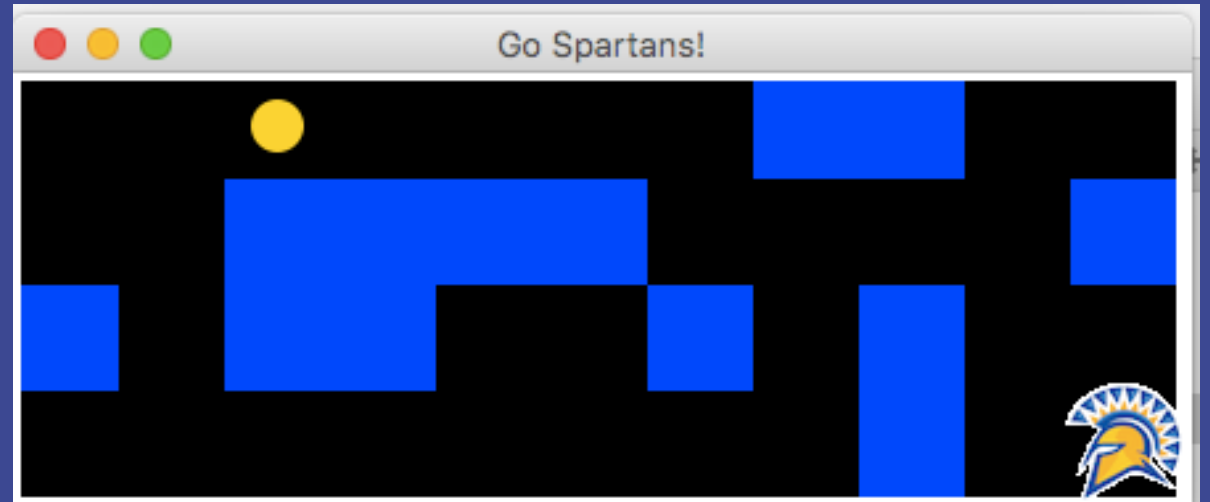
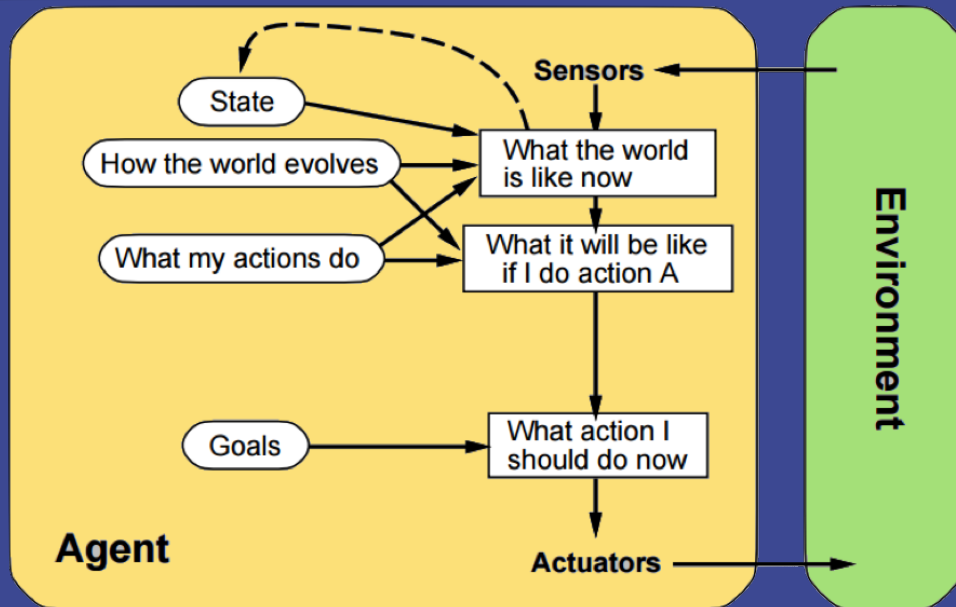


Reflex Agent with State

- ▶ Suppose the vacuum agent remembers the locations visited
- ▶ Suppose it also knows that a clean location stays clean for some time
- ▶ Does not go back to that location for a while

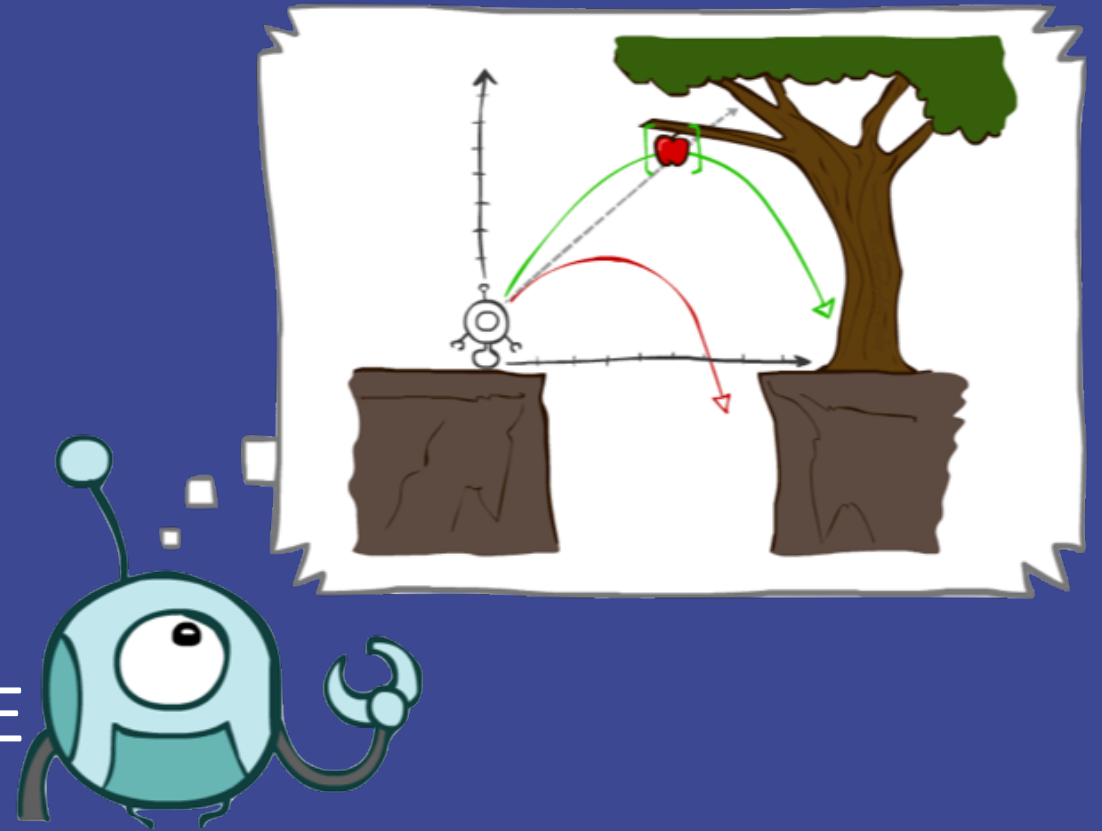


Goal Based Agent



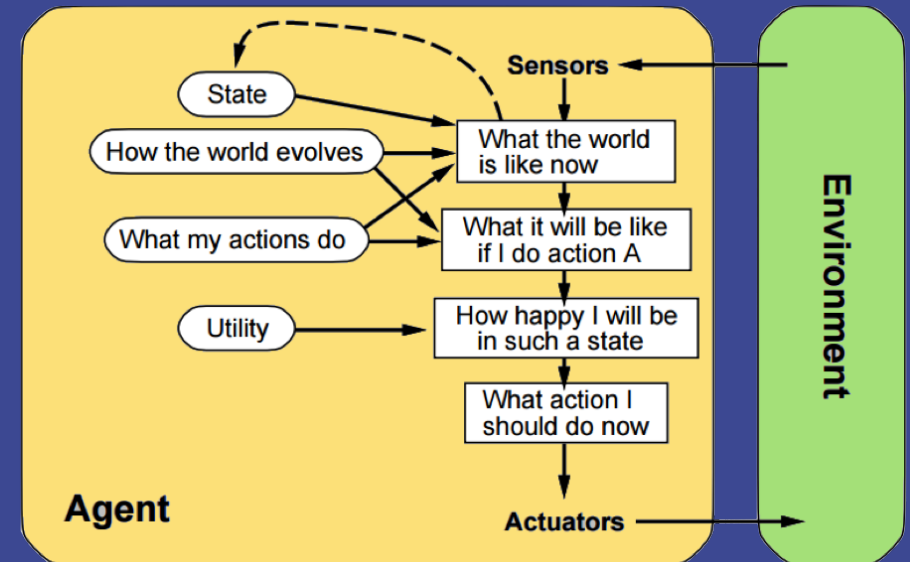
Goal Based Agent

- ▶ Ask "what if"
- ▶ Decisions based on (hypothesized) consequences of actions
- ▶ Must have a model of how the world evolves in response to actions
- ▶ **Must formulate a goal (test)**
- ▶ Consider how the world WOULD BE

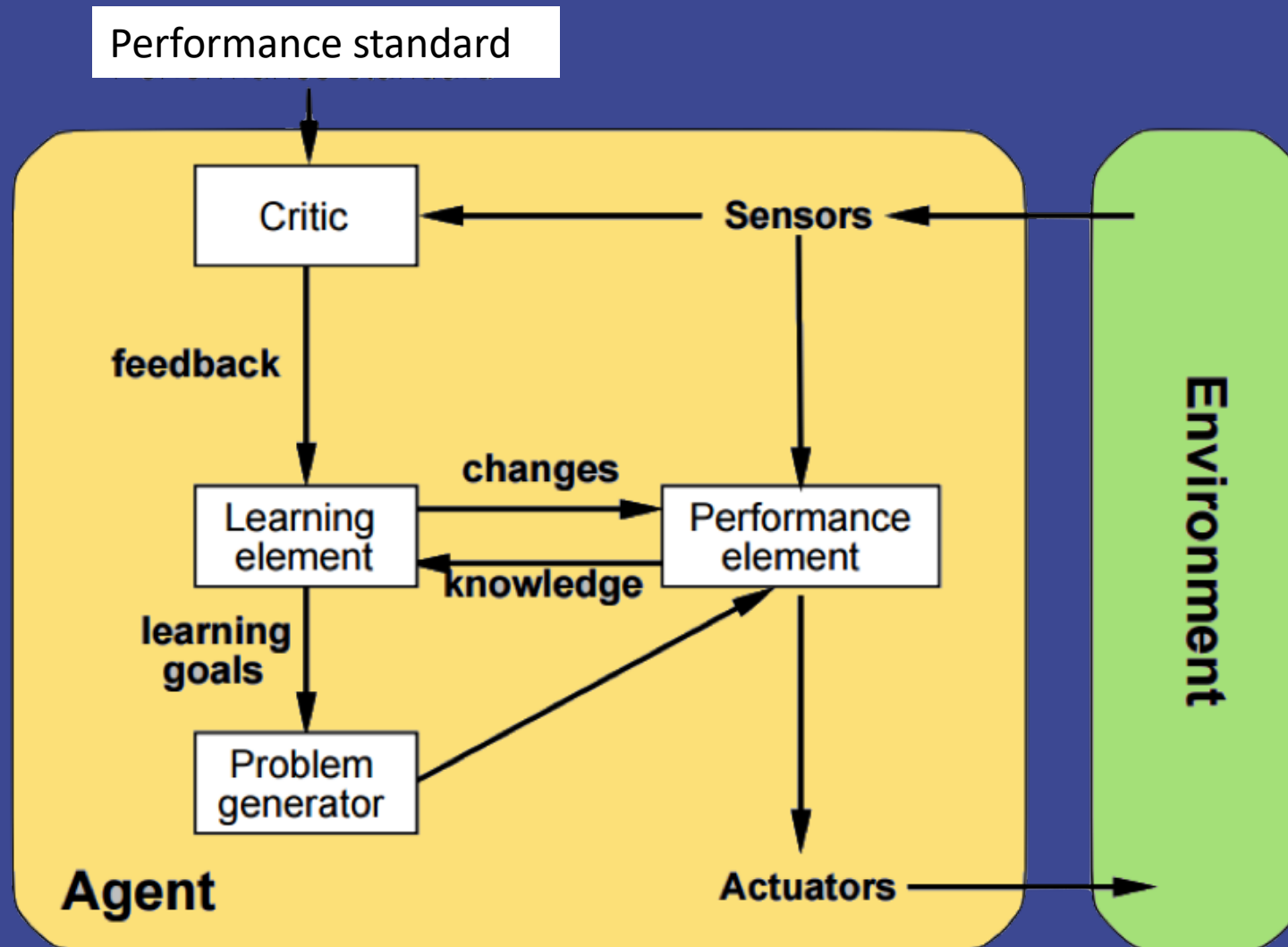


Utility Based Agent

- ▶ Ask "what if"
- ▶ Decisions based on (hypothesized) consequences of actions
- ▶ Must have a model of how the world evolves in response to actions
- ▶ Must formulate a utility - a measure of 'happiness'
- ▶ Select actions that maximize the expected utility
- ▶ Consider how the world WOULD BE



Learning Agent

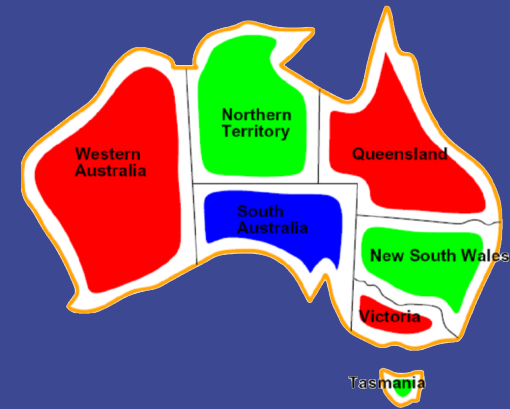
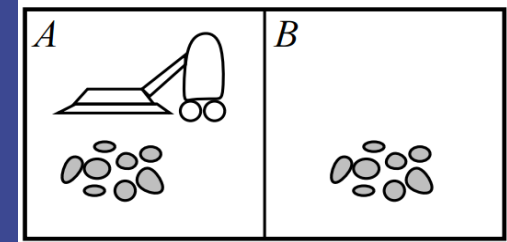


Learning Agent

- ▶ Learning element: responsible for making improvements
- ▶ Performance element: responsible for selecting external actions
- ▶ Critic assesses how well the agent is doing with respect to a fixed performance standard
- ▶ Problem generator: responsible for suggesting exploratory actions

Representing the World

- ▶ Atomic representation: each state of the world is indivisible (black box)
- ▶ Factored representation: states are divided into variables and each variable has a value
- ▶ Structured representation: objects and relationships are described explicitly



Intelligent Agents: Summary

- ▶ An agent perceives and acts in an environment
- ▶ The performance measure evaluates the behavior of the agent
- ▶ A rational agent maximizes its expected performance
- ▶ PEAS descriptions define task environments
- ▶ Environments are categorized along several dimensions: observable, deterministic, episodic, static, discrete, single-agent
- ▶ Basic agent architectures: reflex, reflex with state, goal-based, utility-based
- ▶ Agents can improve their performance through learning

Reminders

- ▶ Question of the Week on Tuesday
- ▶ Bring laptop (with Lockdown browser), cheat sheet
- ▶ Homework 0 - CS 156 Python Essentials due September 8